

customer_shopping_behavior

July 21, 2026

```
[1]: import pandas as pd
import numpy as np
df=pd.read_excel(r"C:\Araib_Projects\customer_shopping_behavior.xlsx")
```

```
[2]: df.head()
```

```
[2]:
```

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	\
0	1	55	Male	Blouse	Clothing	53	
1	2	19	Male	Sweater	Clothing	64	
2	3	50	Male	Jeans	Clothing	73	
3	4	21	Male	Sandals	Footwear	90	
4	5	45	Male	Blouse	Clothing	49	

	Location	Size	Color	Season	Review Rating	Subscription Status	\
0	Kentucky	L	Gray	Winter	3.1	Yes	
1	Maine	L	Maroon	Winter	3.1	Yes	
2	Massachusetts	S	Maroon	Spring	3.1	Yes	
3	Rhode Island	M	Maroon	Spring	3.5	Yes	
4	Oregon	M	Turquoise	Spring	2.7	Yes	

	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	\
0	Express	Yes	Yes	14	
1	Express	Yes	Yes	2	
2	Free Shipping	Yes	Yes	23	
3	Next Day Air	Yes	Yes	49	
4	Free Shipping	Yes	Yes	31	

	Payment Method	Frequency of Purchases
0	Venmo	Fortnightly
1	Cash	Fortnightly
2	Credit Card	Weekly
3	PayPal	Weekly
4	PayPal	Annually

```
[3]: df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
```

Data columns (total 18 columns):

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	str
3	Item Purchased	3900 non-null	str
4	Category	3900 non-null	str
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	str
7	Size	3900 non-null	str
8	Color	3900 non-null	str
9	Season	3900 non-null	str
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	str
12	Shipping Type	3900 non-null	str
13	Discount Applied	3900 non-null	str
14	Promo Code Used	3900 non-null	str
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	str
17	Frequency of Purchases	3900 non-null	str

dtypes: float64(1), int64(4), str(13)

memory usage: 548.6 KB

```
[4]: df.describe(include="all")
```

```
[4]:
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	Customer ID	Age	Gender	Item Purchased	Category	\
count	3900.000000	3900.000000	3900	3900	3900	
unique	NaN	NaN	2	25	4	
top	NaN	NaN	Male	Blouse	Clothing	
freq	NaN	NaN	2652	171	1737	
mean	1950.500000	44.068462	NaN	NaN	NaN	
std	1125.977353	15.207589	NaN	NaN	NaN	
min	1.000000	18.000000	NaN	NaN	NaN	
25%	975.750000	31.000000	NaN	NaN	NaN	
50%	1950.500000	44.000000	NaN	NaN	NaN	
75%	2925.250000	57.000000	NaN	NaN	NaN	
max	3900.000000	70.000000	NaN	NaN	NaN	

	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	\
count	3900.000000	3900	3900	3900	3900	3863.000000	
unique	NaN	50	4	25	4	NaN	
top	NaN	Montana	M	Olive	Spring	NaN	
freq	NaN	96	1755	177	999	NaN	
mean	59.764359	NaN	NaN	NaN	NaN	3.750065	
std	23.685392	NaN	NaN	NaN	NaN	0.716983	
min	20.000000	NaN	NaN	NaN	NaN	2.500000	

25%	39.000000	NaN	NaN	NaN	NaN	3.100000
50%	60.000000	NaN	NaN	NaN	NaN	3.800000
75%	81.000000	NaN	NaN	NaN	NaN	4.400000
max	100.000000	NaN	NaN	NaN	NaN	5.000000

	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	\
count	3900	3900	3900	3900	3900
unique	2	6	2	2	2
top	No	Free Shipping	No	No	No
freq	2847	675	2223	2223	2223
mean	NaN	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN
50%	NaN	NaN	NaN	NaN	NaN
75%	NaN	NaN	NaN	NaN	NaN
max	NaN	NaN	NaN	NaN	NaN

	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900	3900
unique	NaN	6	7
top	NaN	PayPal	Every 3 Months
freq	NaN	677	584
mean	25.351538	NaN	NaN
std	14.447125	NaN	NaN
min	1.000000	NaN	NaN
25%	13.000000	NaN	NaN
50%	25.000000	NaN	NaN
75%	38.000000	NaN	NaN
max	50.000000	NaN	NaN

```
[5]: df.isnull().sum()
```

```
[5]: Customer ID      0
     Age              0
     Gender           0
     Item Purchased    0
     Category          0
     Purchase Amount (USD) 0
     Location          0
     Size              0
     Color             0
     Season            0
     Review Rating     37
     Subscription Status 0
     Shipping Type     0
     Discount Applied  0
```

```
Promo Code Used      0
Previous Purchases    0
Payment Method        0
Frequency of Purchases 0
dtype: int64
```

```
[6]: df["Review Rating"] = df.groupby("Category")["Review Rating"].transform(lambda x:
    ↪x: x.fillna(x.median()))
```

```
[7]: df.isnull().sum()
```

```
[7]: Customer ID      0
Age                  0
Gender               0
Item Purchased       0
Category             0
Purchase Amount (USD) 0
Location             0
Size                 0
Color                0
Season               0
Review Rating        0
Subscription Status  0
Shipping Type        0
Discount Applied     0
Promo Code Used      0
Previous Purchases    0
Payment Method        0
Frequency of Purchases 0
dtype: int64
```

```
[8]: df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

```
[9]: df.columns
```

```
[9]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'promo_code_used', 'previous_purchases',
        'payment_method', 'frequency_of_purchases'],
        dtype='str')
```

```
[ ]: # create a column age_group
df["age_group"] = df["age"].transform(lambda x:
```

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        "Young Adult" if x<=30 else 'Adult' if
↪x<=45 else 'Middle-aged'
        if x<=60 else 'Senior')

```

```
[11]: df[["age", "age_group"]].head(10)
```

```

[11]:   age  age_group
0    55  Middle-aged
1    19  Young Adult
2    50  Middle-aged
3    21  Young Adult
4    45      Adult
5    46  Middle-aged
6    63      Senior
7    27  Young Adult
8    26  Young Adult
9    57  Middle-aged

```

```

[12]: # create a column purchase_frequency_days
frequency_mapping = {
    'Fortnightly': 14,
    'Weekly': 7,
    'Monthly': 30,
    'Quarterly': 90,
    'Bi-Weekly': 14,
    'Annually': 365,
    'Every 3 Months': 90
}
df["purchase_frequency_days"] = df["frequency_of_purchases"].
↪map(frequency_mapping)

```

```
[13]: df[["purchase_frequency_days", "frequency_of_purchases"]].head(100)
```

```

[13]:   purchase_frequency_days  frequency_of_purchases
0                14      Fortnightly
1                14      Fortnightly
2                 7        Weekly
3                 7        Weekly
4               365      Annually
..              ...
95               30        Monthly
96               90  Every 3 Months
97               365      Annually
98               365      Annually
99               90      Quarterly

```

[100 rows x 2 columns]

```
[14]: df[["discount_applied", "promo_code_used"]].head(10)
```

```
[14]: discount_applied promo_code_used
0          Yes          Yes
1          Yes          Yes
2          Yes          Yes
3          Yes          Yes
4          Yes          Yes
5          Yes          Yes
6          Yes          Yes
7          Yes          Yes
8          Yes          Yes
9          Yes          Yes
```

```
[15]: (df["discount_applied"] == df["promo_code_used"]).all()
```

```
[15]: np.True_
```

```
[16]: df = df.drop("promo_code_used", axis=1)
```

```
[17]: df.columns
```

```
[17]: Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
        'purchase_amount', 'location', 'size', 'color', 'season',
        'review_rating', 'subscription_status', 'shipping_type',
        'discount_applied', 'previous_purchases', 'payment_method',
        'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],
        dtype='str')
```

! pip install psycpg2-binary sqlalchemy

! pip install pymysql

```
[ ]: import pandas as pd
from sqlalchemy import create_engine

# my sql connection detail
username = "root"
password = "malik2000#"
host = "127.0.0.1"
port = "3306"
database = "customer_behavior"

# create mysql connection
engine = create_engine(f"mysql+pymysql://{username}:{password}@{host}:{port}/
↳ {database}")
print("my sql connected successfully")
```

```
# upload dataframe to my sql  
df.to_sql(name="customer_data", con=engine, if_exists="replace", index=False)  
print("data uploaded successfully")
```

```
[ ]:
```